6. Conclusions and outlook

We performed comprehensive investigations of the energy dependence of aberration-free plasma generation in water by tightly focused laser pulses that cover a wide range of pulse durations (fs to ns), wavelengths (UV to IR), and numerical apertures. Deterministic behavior was observed not only in fs breakdown but also, when UV- or VIS ns pulses with smooth, reproducible temporal shape are tightly focused into pure dielectrics. This extends the basis for parameter selection in laser surgery and material processing.

We established a conceptual framework of nonlinear energy deposition in (τ_L, λ_L, NA, E_L) parameter space by analyzing the interplay between PI, AI, recombination, and thermal ionization. On this basis, we identified three different scenarios of nonlinear energy deposition and confirmed their existence through experimental investigations. The scenarios are: 1. *Tunable energy deposition by ultrashort laser pulses of all wavelengths, with continuous transition from nanoeffects to extreme energy densities and disruptive effects*. The possibility of nanoeffects relies on the abundance of multiphoton-mediated seed electrons for AI already at the breakdown threshold, and the smooth tunability is due to time constraints of AI. 2. *Big effects already at threshold in IR ns breakdown.* Here, the generation of seed electrons by MPI is the critical hurdle for breakdown and, due to the high order of multiphoton effects, this hurdle is overcome only at high irradiance values, where AI largely overshoots the bubble threshold. 3. *Stepwise transition from nanoeffects close to threshold to extreme energy densities well above threshold in UV and VIS ns breakdown*. Here, seed electrons are readily available due to the low order of MPI, and AI is close to the bubble threshold inhibited by recombination, which provides fine-tunability of energy deposition. Above a second threshold, the temperature in the focal volume is so high that thermal ionization sets in during the laser pulse. Now, AI and TI jointly overcome the inhibition by recombination and suddenly high degrees of ionization are reached.

On a mesoscopic level, ionization involves an interplay between downstream or upstream movement of the breakdown wave, spatiotemporal instabilities of free-electron multiplication by AI, and X-ray-mediated radiative energy transport from high-density strings or hot spots into their surroundings. The combination of interplays on a molecular and a mesoscopic level results in a higher complexity of optical breakdown events, especially for ns plasmas, than previously envisioned.

The plasma energy density in bulk media is limited by the movement of the breakdown front during the laser pulse. The downstream movement of the breakdown wave in fs breakdown at small to moderate

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NA in conjunction with nonlinear beam propagation results in energy depletion during pulse propagation and low electron and energy density. By contrast, high energy densities are produced at large *NA*, when the plasma is shorter than the laser pulse and the breakdown front moves upstream. The largest average plasma energy densities reach up to $U_{avg} \approx 40$ kJ/cm^3 at $NA = 0.8$ both for fs and ns breakdown. This corresponds to peak pressures above 10 GPa. The local energy density of ns plasmas can reach much higher values. String-like regions of primary energy deposition during optical breakdown at large *NA* have an estimated energy density of up to 400 kJ/cm^3, and the estimated peak value in the hot spot observed in the upstream part of the plasma at $NA = 0.3$ is 260 kJ/cm^3, corresponding to a pressure above 100 GPa.

The $R_{max}(E_L)$ curves obtained at different pulse durations, wavelengths, and *NA*s, characterize the different scenarios and delineate regimes with low and high plasma energy density. The $R_{max}(E_L)$ curves also show the existence of a general scaling law for high-density plasmas, regardless of pulse duration and wavelength. In this regime, well above the respective breakdown thresholds, the bubble volume and energy are proportional to the incident laser pulse energy. The partitioning of absorbed laser energy into vaporization-, bubble-, and shock wave energy was explored as a function of plasma energy density in a range between 8.5 kJ/cm^3 and 40 kJ/cm^3. With increasing energy density, the conversion into mechanical energy grows rapidly up to about 15 kJ/cm^3 and more gradually for larger plasma energy densities. While the fraction going into bubble energy stays approximately constant near 20%, the fraction needed for vaporization drops to only 7% at $U_{avg} = 40$ kJ/cm^3, while more than ¾ of the absorbed energy is converted into shock wave energy.

The discovery that precisely tunable energy deposition is possible in larger regions of the (τ_L, λ_L,) space than previously thought provides new possibilities for cost-effective and compact solutions adapted to specific applications. We demonstrated the utility of tunable nonlinear energy deposition with UV ns pulses on the examples of refractive corneal surgery as well as on micro-marking and refractive index modifications in glass. These examples indicate that results for water obtained in this study are representative for breakdown features in many transparent dielectrics.

The identification of different scenarios through systematic investigation of $R_{max}(E_L)$ and the exploration of energy partitioning of absorbed laser energy into bubble and shock wave energy demonstrated in this paper on breakdown in bulk liquid can be useful also for delineating regimes with different working mechanisms in LAL and LIBS. We found that scenarios 1-3 are most clearly separated,

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when nanosecond laser pulses with smooth temporal shape are employed rather than the common multi-longitudinal-mode pulses exhibiting intensity spikes from mode beating. Therefore, it seems worthwhile to investigate the influence of the mode of laser operation also in LAL and LIBS.

6.1. Roadmap for future experimental work

The experimental results in this paper were obtained by probe beam scattering that enabled to collect a large amount of $R_{max}(E_L)$ data in the (τ_L, λ, NA) parameter space, and by time-integrated high-resolution plasma photography in side view together with transmission measurements. Evaluation of plasma volume and transmission yielded the plasma energy density. The color photographs made is possible to distinguish between regions of primary energy deposition and radiative plasma enlargement and helped to discriminate bremsstrahlung and blackbody radiation. Information on the breakdown dynamics was inferred with the help of the moving breakdown model. An extension of high-resolution color photography to plasma formation at solid targets may provide a deeper understanding also for the mechanisms of LAL and LIBS.

Further exploration of the breakdown dynamics both in bulk media and at solid targets requires time resolved investigations of the sequence of breakdown events with high spatiotemporal resolution. Time-resolved ICCD imaging with ultrashort gating times can directly portray the plasma growth during nanosecond breakdown and visualize the formation of branched high-density strings and of large hot spots during the movement of the breakdown wave. Shadowgraph and schlieren photography can provide insights into the pressure distribution inside and around the plasma and the resulting shock wave formation. Presumably, the local hydrodynamic expansion of string regions and hot spots featuring high energy density and pressure will be faster than the expansion of the larger region emitting diffuse plasma luminescence in which energy density and pressure are lower. The difference has to date been obscured because tracking of acoustic transients and shock wave propagation was restricted to the plasma surroundings. With sufficiently bright narrowband light sources, picosecond exposure times and sub-micrometer image resolution, the propagation of shock waves emanating from hot spots or strings can possibly be tracked also inside the diffusely luminescent region.

Time-resolved spectroscopy can provide information on the transition from the initial plasma state to a vapor bubble during the initial expansion phase. Blackbody temperatures of the breakdown plasma are available [224-226], and spectroscopic temperatures from line spectra are also available for $t \geq 75$ ns [103] but the transition from continuum emission to line emission has not yet been continuously tracked. Coverage of this phase with spectroscopic measurements of plasma temperature and electron density

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